Holo pad

Nova HoloPad — System Architecture

1. Main Computing System

The Raspberry Pi 5 is planned as the central computing unit.
Responsibilities:

- Run the Nova assistant.
- Process user commands.
- Manage display interaction.
- Process webcam input.
- Communicate with motor controllers.

2. Display System

A 55-inch touchscreen will provide the primary visual interface.

Planned functions:

- Display Nova's interface.
- Receive touch input.
- Show system information.
- Display interactive applications.

3. Vision System

A webcam will provide visual input.
Planned functions:

- Capture video.
- Detect hand landmarks.
- Recognise supported gestures.
- Send recognised commands to Nova.

4. Motion System

Motorised mechanisms are planned for display movement.

The motor-control subsystem will operate independently from the main computing unit, with appropriate electrical and mechanical safety limits.

5. Software Architecture

Input → Gesture Recognition / Voice Recognition → Nova → Intent Processing → Output
Outputs:

- Touchscreen interface
- Voice responses
- Approved motor-control commands

6. Current Development Status

This document describes the proposed architecture. Hardware integration and physical testing are pending funding.